

Size	Method	Live		Non-Live		Multi-Turn		Average	
		Acc.	Format	Acc.	Format	Acc.	Format	Acc.	Format
1.5B	Base	9.33	0.37	12.06	0.12	0.38	0.21	7.25	0.23
	GRPO	64.80 $\pm$ 0.55	99.88 $\pm$ 0.04	78.04 $\pm$ 0.84	99.91 $\pm$ 0.12	2.55 $\pm$ 0.46	43.24 $\pm$ 15.52	48.46 $\pm$ 0.36	81.01 $\pm$ 5.18
	GDPO	67.05 $\pm$ 2.40	99.94 $\pm$ 0.06	79.30 $\pm$ 1.69	99.97 $\pm$ 0.06	4.75 $\pm$ 0.74	91.40 $\pm$ 3.02	50.37 $\pm$ 1.12	97.11 $\pm$ 1.00
	DVAO	64.99 $\pm$ 4.64	99.75 $\pm$ 0.23	76.84 $\pm$ 3.30	99.62 $\pm$ 0.45	4.08 $\pm$ 0.14	81.19 $\pm$ 8.28	48.64 $\pm$ 2.64	93.52 $\pm$ 2.91
	GD ² PO-Hard	67.97 $\pm$ 1.56	99.98 $\pm$ 0.04	79.44 $\pm$ 1.01	100.00 $\pm$ 0.00	4.67 $\pm$ 1.05	89.08 $\pm$ 2.16	50.70 $\pm$ 0.98	96.35 $\pm$ 0.73
	DARA	68.42 $\pm$ 1.31	99.91 $\pm$ 0.12	80.25 $\pm$ 0.78	99.97 $\pm$ 0.06	4.85 $\pm$ 0.68	92.77 $\pm$ 1.88	51.17 $\pm$ 0.78	97.55 $\pm$ 0.64
3B	Base	47.67	54.92	36.23	34.23	1.75	9.16	28.55	32.77
	GRPO	69.31 $\pm$ 0.25	99.78 $\pm$ 0.12	81.86 $\pm$ 0.82	99.84 $\pm$ 0.19	6.53 $\pm$ 0.83	66.66 $\pm$ 3.63	52.57 $\pm$ 0.44	88.76 $\pm$ 1.22
	GDPO	69.30 $\pm$ 0.99	99.93 $\pm$ 0.09	82.60 $\pm$ 0.94	100.00 $\pm$ 0.00	9.07 $\pm$ 1.49	89.79 $\pm$ 2.34	53.66 $\pm$ 0.96	96.57 $\pm$ 0.77
	DVAO	70.00 $\pm$ 0.56	99.90 $\pm$ 0.09	83.03 $\pm$ 0.50	99.99 $\pm$ 0.01	9.62 $\pm$ 0.33	91.21 $\pm$ 1.49	54.22 $\pm$ 0.24	97.04 $\pm$ 0.47
	GD ² PO-Hard	69.45 $\pm$ 1.58	99.73 $\pm$ 0.04	83.11 $\pm$ 1.37	100.00 $\pm$ 0.00	9.12 $\pm$ 1.21	91.28 $\pm$ 2.45	53.90 $\pm$ 1.11	97.00 $\pm$ 0.82
	DARA	70.39 $\pm$ 0.98	99.97 $\pm$ 0.04	83.05 $\pm$ 0.81	100.00 $\pm$ 0.00	9.00 $\pm$ 0.99	91.69 $\pm$ 2.02	54.15 $\pm$ 0.58	97.22 $\pm$ 0.66

BFCL V4 · step 100 · mean $\pm$ sample SD (%)

GRPO / GDPO / DARA: 5 seeds per size; DVAO / GD²PO-Hard: 3 retained seeds per size.

Retention: step-100 RLLA validation Format $\geq$ 80%. Average = mean over Live, Non-Live, and Multi-Turn.